Project 1: Compute Word Frequencies in a File

Introduction

In this project, I created a C++ program to compute the frequency of words in a text file. The program reads a file named "holmes.txt" and counts how many times each word appears. The goal of this project is to practice using vectors, strings, loops, and functions in C++.

Requirements / Scope

The requirements of this project were:

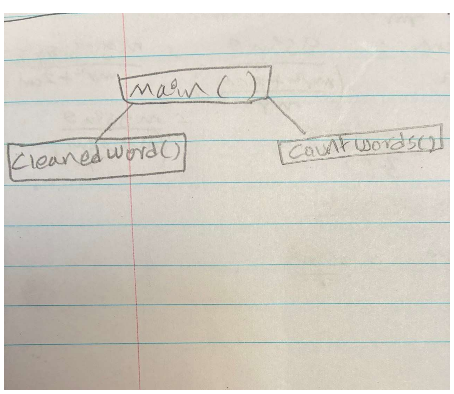
- Read a text file
- Store words in a vector
- Count word frequency
- Use functions to make the program modular
- Display the results

The scope of this project includes reading the file, cleaning words, counting frequencies, and printing the output.

Design

The program is divided into the following functions:

- main()
- cleanedword()
- countWords()



The main function reads the file and stores the words. The cleanedword function removes punctuation from words. The countWords function counts the frequency of each word.

Development

GitHub Repository URL: [Fullreton-College/project-1-nigist-02083466: fullreton-college-classroom-24510-project-1-fullreton-college-csci123-100f-project-1-word-frequencie created by GitHub Classroom]

Code Name: project-1.cpp

Did you get any help from AI?
Yes, I received guidance to understand the structure of the program and debugging help. However, I wrote and tested the code myself.

Screenshots

project-1.cpp > cleanedword(string)

```cpp
#include <iostream>
#include <fstream>
#include <string>
#include <vector>

using namespace std;

vector<string> words;
vector<string> uniqueWords;
vector<int> counts;

// clean word function
string cleanedword(string w)
{
    string clean = "";

    for (char c : w)
    {
        if ((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z'))
        {
            clean += c;
        }
    }

    return clean;
}

```

```cpp
// count words
void countWords()
{
    for (int i = 0; i < words.size(); i++)
    {
        bool found = false;

        for (int j = 0; j < uniqueWords.size(); j++)
        {
            if (words[i] == uniqueWords[j])
            {
                counts[j]++;
                found = true;
                break;
            }
        }

        if (!found && words[i] != "")
        {
            uniqueWords.push_back(words[i]);
            counts.push_back(1);
        }
    }
}
```

```cpp
int main()
{
    ifstream file("homes.txt");

    if (!file.is_open())
    {
        cout << "File is not open" << endl;
        return 1;
    }

    string word;

    while (file >> word)
    {
        word = cleanedword(word);
        words.push_back(word);
    }

    countWords();

    for (int i = 0; i < uniqueWords.size(); i++)
    {
        cout << uniqueWords[i] << " : " << counts[i] << endl;
    }

    file.close();
}
```

Output

```
PS C:\Users\Nigist Tekle\Desktop\Project 1 C++> g++ -o project-1.exe project-1.cpp
PS C:\Users\Nigist Tekle\Desktop\Project 1 C++> ./project-1.exe
To : 1
Sherlock : 1
Holmes : 1
she : 2
is : 1
always : 1
the : 9
woman : 3
I : 2
have : 2
seldom : 1
heard : 1
him : 2
mention : 1
her : 2
under : 1
any : 2
other : 1
name : 1
In : 1
his : 6
eyes : 1
eclipses : 1
and : 8
predominates : 1
whole : 1
of : 4
sex : 1
It : 1
was : 5
not : 2
that : 4
he : 2
felt : 1
emotion : 2
akin : 1
to : 5
love : 1
for : 4
```

```
Irene : 2
Adler : 2
All : 1
emotions : 1
one : 3
particularly : 1
were : 2
abhorrent : 1
cold : 1
precise : 1
but : 3
admirably : 1
balanced : 1
mind : 1
He : 2
take : 1
it : 1
most : 1
perfect : 1
reasoning : 1
observing : 1
machine : 1
world : 1
has : 1
seen : 1
as : 2
a : 10
lover : 1
would : 2
placed : 1
himself : 1
in : 4
false : 1
position : 1
never : 1
spoke : 1
softer : 1
passions : 1
save : 1
with : 1
```

```
gibe : 1
sneer : 1
They : 1
admirable : 1
things : 1
observerexcellent : 1
drawing : 1
veil : 1
from : 1
mens : 1
motives : 1
actions : 1
But : 1
trained : 1
reasoner : 1
admit : 1
such : 2
intrusions : 1
into : 1
own : 2
delicate : 1
finely : 1
adjusted : 1
temperament : 1
introduce : 1
distracting : 1
factor : 1
which : 1
might : 1
throw : 1
doubt : 1
upon : 1
all : 1
mental : 1
results : 1
Grit : 1
sensitive : 1
instrument : 1
or : 1
crack : 1
highpower : 1
```

```
lenses : 1
be : 1
more : 1
disturbing : 1
than : 1
strong : 1
nature : 1
And : 1
yet : 1
there : 1
late : 1
dubious : 1
questionable : 1
memory : 1
PS C:\Users\Nigist Tekle\Desktop\Project 1 C++>
```

Conclusion

In this project, I learned how to read files, use vectors, and count word frequencies in C++. I also learned how to divide a program into functions. This project helped me improve my understanding of C++ programming and problem solving.